

Highlighted Manuscript: Change and Deletion Index

Baseline: 33f5e3d37a85641b13bffaf7fc34b94eb5d35c47. Selected as the latest manuscript before August 13 under the author's instruction. Yellow marks show inserted/replaced rendered words; orange frames identify pages containing changed figure assets. The clean PDF text is unchanged by annotations. Source-only layout/preamble edits are preserved in reviewed-baseline-source.diff. This index records removed text that cannot be highlighted in the revised manuscript.

Change 1: replace | old pp. [1] | new pp. [1]

Before (including deletions): SPICE

After: Structural Circuit

Change 2: delete | old pp. [1] | new pp. []

Before (including deletions): Automated digitization of hand-drawn circuit schematics into simulation-ready SPICE netlists remains challenging because connectivity recovery is brittle under noisy wire detections.

After: (none)

Change 3: replace | old pp. [1] | new pp. [1]

Before (including deletions): that converts a scanned schematic into a SPICE netlist through component detection,

After: for structural circuit-netlist recovery using

Change 4: replace | old pp. [1] | new pp. [1]

Before (including deletions): occlusion-first

After: occlusion- first

Change 5: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: an endpoint graph,

Change 6: replace | old pp. [1] | new pp. [1]

Before (including deletions): a graph-based join that resolves electrical connectivity.

After: degree-budget completion.

Change 7: replace | old pp. [1] | new pp. [1]

Before (including deletions): main contribution

After: output represents inferred pin-to-node assignments; our component-pair metric does not certify pin identities, exact net partitions, absence of shorts, or simulation equivalence. Illustrative SPICE export requires externally specified values and device models; simulation equivalence on real scans

Change 8: replace | old pp. [1] | new pp. [1]

Before (including deletions): the joining stage: an endpoint-graph model over wire endpoints and component pins with five edge types, combined with degree-budget completion that recovers dropped connections through min- cost b-matching while bounding over-merge with per-pin budgets and self-loop guards.

After: not validated.

Change 9: replace | old pp. [1] | new pp. [1]

Before (including deletions): also introduce

After: provide

Change 10: replace | old pp. [1] | new pp. [1]

Before (including deletions): and, to our knowledge, the first

After: and a

Change 11: delete | old pp. [1] | new pp. []

Before (including deletions): net-level

After: (none)

Change 12: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: of 31 CGHD-1152 images. With annotated component boxes and detected wires, the join achieves component-pair micro-F1 0.890 (macro- F1 0.901), compared with 0.805

Change 13: replace | old pp. [1] | new pp. [1]

Before (including deletions): hand-drawn circuits, covering 31 images from CGHD-1152.

After: Hough plus proximity and 0.624 for connected-component tracing on the same detected wires.

Change 14: replace | old pp. [1] | new pp. [1]

Before (including deletions): occlusion-first

After: wire

Change 15: replace | old pp. [1] | new pp. [1]

Before (including deletions): reaches wire-detection

After: achieves

Change 16: delete | old pp. [1] | new pp. []

Before (including deletions): =

After: (none)

Change 17: replace | old pp. [1] | new pp. [1]

Before (including deletions): scanned images. On

After: images with annotated component boxes under

Change 18: replace | old pp. [1] | new pp. [1]

Before (including deletions): human-verified net-level

After: historical benchmark configuration. Under the same component priors on the 31-image

Change 19: delete | old pp. [1] | new pp. []

Before (including deletions): the best join reaches connectivity micro-F1 = 0.890, outperforming the prior completion default (0.83),

After: (none)

Change 20: delete | old pp. [1] | new pp. []

Before (including deletions): legacy radius union-find join (0.67), a connected-component net-tracing baseline (0.62), and a Hough- plus-proximity line tracer (0.81), all under identical detected wires. Running the join on perfect wire labels does not improve micro-F1, showing that connectivity, not wire detection, remains the bottleneck. A first- party comparison shows a strong

After: (none)

Change 21: replace | old pp. [1] | new pp. [1]

Before (including deletions): 0.923 on

After: micro-F1 0.923;

Change 22: replace | old pp. [1] | new pp. [1]

Before (including deletions): same images, statistically indistinguishable from our join but far more expensive and without structural guarantees, supporting explicit geometric connectivity recovery for deployable circuit digitization.

After: paired difference is +0.033, with 95% confidence interval $[-0.009, +0.078]$. Inclusion of zero does not establish equivalence or comparative cost superiority. These results provide within-corpus evidence, not validation of autonomous reconstruction or unseen-style generalization.

Change 23: replace | old pp. [1] | new pp. [1]

Before (including deletions): cir- cuits,

After: circuits,

Change 24: replace | old pp. [1] | new pp. [1]

Before (including deletions): functional information (component

After: component

Change 25: replace | old pp. [1] | new pp. [1]

Before (including deletions): connections)

After: connections

Change 26: replace | old pp. [1] | new pp. [1]

Before (including deletions): Converting scanned schematics into machine-readable

After: Recovering a structural netlist from a scan is a prerequisite for simulation, design verification, and large-scale circuit analysis, but a complete simulatable

Change 27: replace | old pp. [1] | new pp. [1]

Before (including deletions): (Simula- tion

After: (Simulation

Change 28: replace | old pp. [1] | new pp. [1]

Before (including deletions): netlists en- ables automated simulation, design verification,

After: deck also requires component values

Change 29: replace | old pp. [1] | new pp. [1]

Before (including deletions): large- scale circuit analysis. While recent advances in object detec- tion have made component recognition increasingly reliable,

After: device models. This work addresses

Change 30: insert | old pp. [] | new pp. [1]

Before (including deletions): (none)

After: structural connectivity stage only. The

Change 31: replace | old pp. [1] | new pp. [1]

Before (including deletions): problem, determining

After: problem determines

Change 32: replace | old pp. [1] | new pp. [1]

Before (including deletions): con- nect

After: connect

Change 33: replace | old pp. [1] | new pp. [1]

Before (including deletions): other, remains the primary failure mode in end-to-end digitization pipelines.

After: other. We study this stage with annotated component boxes; its relative contribution to autonomous reconstruction error is not established by that experiment.

Change 34: replace | old pp. [1] | new pp. [1]

Before (including deletions): fragments,

After: fragments,

Change 35: delete | old pp. [1] | new pp. []

Before (including deletions): complex

After: (none)

Change 36: replace | old pp. [1] | new pp. [1]

Before (including deletions): patterns that simple radius-based approaches cannot resolve without over-merging.

After: ambiguities for proximity-based rules, including the graph rules studied here. A complementary approach is to prompt a vision-language model (VLM) for connectivity. Section IV-G reports an oracle-component-box diagnostic in which both methods receive the same annotated component boxes and the geometric method uses detected wires. The observed component-pair

Change 37: replace | old pp. [2] | new pp. [2]

Before (including deletions): SPICE

After: Structural

Change 38: replace | old pp. [2] | new pp. [2]

Before (including deletions): Pin discovery + simulation

After: Inferred pin-to-node map

Change 39: replace | old pp. [2] | new pp. [2]

Before (including deletions): End-to-end pipeline

After: Pipeline

Change 40: replace | old pp. [2] | new pp. [2]

Before (including deletions): scanning circuit diagram digitization. Our contributions span the

After: structural circuit-netlist recovery. The highlighted stages extract

Change 41: replace | old pp. [2] | new pp. [2]

Before (including deletions): extraction

After: endpoints

Change 42: replace | old pp. [2] | new pp. [2]

Before (including deletions): joining stages; the join (dashed red) determines which extracted wire segments connect

After: infer connectivity. The net-level experiments use annotated component boxes

Change 43: replace | old pp. [2] | new pp. [2]

Before (including deletions): which component pins and is the dominant failure mode in

After: isolate joining; they do not establish an

Change 44: replace | old pp. [2] | new pp. [2]

Before (including deletions): digitization.

After: error hierarchy or validate illustrative SPICE export.

Change 45: replace | old pp. [2] | new pp. [2]

Before (including deletions): (36

After: (35

Change 46: replace | old pp. [2] | new pp. [2]

Before (including deletions): 12

After: 7

Change 47: replace | old pp. [2] | new pp. [2]

Before (including deletions): (12

After: (11

Change 48: replace | old pp. [2] | new pp. [2]

Before (including deletions): A natural alternative is to prompt

After: scores do not establish equivalence, comparative cost superiority or

Change 49: replace | old pp. [2] | new pp. [2]

Before (including deletions): large vision-language model (VLM) to read

After: global structural-validity guarantee. The diagnostic examines

Change 50: replace | old pp. [2] | new pp. [2]

Before (including deletions): schematic and emit its netlist directly. We test this in Section IV-F: a strong VLM attains connectivity accuracy statistically indistinguishable from our geometric pipeline, but it does so at orders-of-magnitude higher cost, returns free-form text rather than a structured netlist, offers no structural guarantee against electrically invalid output (e.g. shorted two-terminal parts), and drops connections on the most complex circuits. These properties motivate an explicit, inspectable geometric pipeline whose output is a netlist by construction and whose accuracy matches the VLM's while being deterministic and simulatable.

After: methods under those stated input conditions.

Change 51: delete | old pp. [2] | new pp. []

Before (including deletions): complete

After: (none)

Change 52: insert | old pp. [] | new pp. [2]

Before (including deletions): (none)

After: structural-netlist

Change 53: replace | old pp. [2] | new pp. [2]

Before (including deletions): novel

After: proposed

Change 54: replace | old pp. [2] | new pp. [2]

Before (including deletions): binarization

After: binarization

Change 55: replace | old pp. [2] | new pp. [2]

Before (including deletions): line- segment-with-endpoints representation

After: line-segment-with-endpoints repre- sentation

Change 56: replace | old pp. [2] | new pp. [2]

Before (including deletions): graph-based

After: graph- based

Change 57: replace | old pp. [2] | new pp. [2]

Before (including deletions): cap- turing

After: capturing

Change 58: replace | old pp. [2] | new pp. [2]

Before (including deletions): identifies and reconnects dropped

After: proposes recovery

Change 59: replace | old pp. [2] | new pp. [2]

Before (including deletions): constrained optimization.

After: bounded assignment and merge guards.

Change 60: delete | old pp. [2] | new pp. []

Before (including deletions): both

After: (none)

Change 61: replace | old pp. [2] | new pp. [2]

Before (including deletions): injec- tion

After: injection

Change 62: delete | old pp. [2] | new pp. []

Before (including deletions): that enables systematic robustness analysis

After: (none)

Change 63: replace | old pp. [2] | new pp. [2]

Before (including deletions): human-verified

After: human- verified

Change 64: insert | old pp. [] | new pp. [2]

Before (including deletions): (none)

After: We evaluate topological recov- ery with illustrative SPICE export; component values and device models require external specification, and simulation equivalence on real scans is not validated. The output records inferred pin-to-node assignments, whereas the primary metric measures component-pair connectivity rather than exact pin topology.

Change 65: replace | old pp. [2] | new pp. [2]

Before (including deletions): connec- tivity,

After: connectiv- ity,

Change 66: replace | old pp. [2] | new pp. [2]

Before (including deletions): (Auto-SPICE)

After: (Auto- SPICE)

Change 67: replace | old pp. [2] | new pp. [2]

Before (including deletions): com- parable

After: comparable

Change 68: replace | old pp. [2] | new pp. [2]

Before (including deletions): image-to- SPICE

After: image-to-SPICE

Change 69: replace | old pp. [2] | new pp. [2]

Before (including deletions): node-detection

After: node- detection

Change 70: replace | old pp. [2] | new pp. [2]

Before (including deletions): and, as we do,

After: and

Change 71: replace | old pp. [2, 3] | new pp. [2]

Before (including deletions): simula- 2 VOLUME 11, 2023 tion

After: simulation

Change 72: insert | old pp. [] | new pp. [2]

Before (including deletions): (none)

After: 2 VOLUME 11, 2023

Change 73: replace | old pp. [3] | new pp. [3]

Before (including deletions): whole-netlist

After: whole- netlist

Change 74: replace | old pp. [3] | new pp. [3]

Before (including deletions): baseline

After: base- line

Change 75: replace | old pp. [3] | new pp. [3]

Before (including deletions): (LT- spice

After: (LTspice

Change 76: replace | old pp. [3] | new pp. [3]

Before (including deletions): com- ponents

After: components

Change 77: replace | old pp. [3] | new pp. [3]

Before (including deletions): CGHD-1152,

After: CGHD- 1152,

Change 78: replace | old pp. [3] | new pp. [3]

Before (including deletions): comparable, but the

After: comparable. Our controlled

Change 79: replace | old pp. [3] | new pp. [3]

Before (including deletions): shows contour/CCL node detection remains the prevailing approach, which our endpoint-graph join substantially outperforms

After: evaluates a CCL baseline

Change 80: replace | old pp. [3] | new pp. [3]

Before (including deletions): a controlled, identical-input benchmark. Instance-segmentation

After: identical detected wires, rather than reproducing their complete system. Instance-segmentation

Change 81: replace | old pp. [3] | new pp. [3]

Before (including deletions): [6]

After: [8]

Change 82: replace | old pp. [3] | new pp. [3]

Before (including deletions): extraction

After: extraction

Change 83: replace | old pp. [3] | new pp. [3]

Before (including deletions): inputs,

After: inputs,

Change 84: replace | old pp. [3] | new pp. [3]

Before (including deletions): [7]

After: [9]

Change 85: replace | old pp. [3] | new pp. [3]

Before (including deletions): JUHCCR-v1

After: JUHCCR-v1

Change 86: replace | old pp. [3] | new pp. [3]

Before (including deletions): [8]

After: [10]

Change 87: replace | old pp. [3] | new pp. [3]

Before (including deletions): [9], [10]

After: [11], [12]

Change 88: replace | old pp. [3] | new pp. [3]

Before (including deletions): [11],

After: [13],

Change 89: replace | old pp. [3] | new pp. [3]

Before (including deletions): [12].

After: [14].

Change 90: replace | old pp. [3] | new pp. [3]

Before (including deletions): establish

After: address detection, recognition and connectivity under different datasets and metrics. Our contribution evaluates joining under specified component priors, without treating

Change 91: replace | old pp. [3] | new pp. [3]

Before (including deletions): and

After: or

Change 92: replace | old pp. [3] | new pp. [3]

Before (including deletions): largely solved; the bottleneck shifts to reconstructing electrical connectivity, the focus of our contribution.

After: solved. SPICE-level circuit modeling likewise underpins application deployment in neighboring hardware domains, including memristor-based neuromorphic circuits that realize brain- inspired learning in hardware [6] and memristive decision-making frameworks for autonomous navigation [7].

Change 93: replace | old pp. [3] | new pp. [3]

Before (including deletions): [13]

After: [15]

Change 94: replace | old pp. [3] | new pp. [3]

Before (including deletions): [14]

After: [16]

Change 95: replace | old pp. [3] | new pp. [3]

Before (including deletions): [15]

After: [17]

Change 96: replace | old pp. [3] | new pp. [3]

Before (including deletions): We use these

After: These

Change 97: replace | old pp. [3] | new pp. [3]

Before (including deletions): both as evidence

After: provide context for component recognition and connectivity evaluation. Our synthetic perturbation protocol isolates joining under authored errors; it does not measure the relative autonomous error contributions

Change 98: replace | old pp. [3] | new pp. [3]

Before (including deletions): where current systems succeed (component recognition) and where they fail (connectivity), and contribute a synthetic robustness protocol that isolates the join step under controlled detector noise.

After: those stages.

Change 99: replace | old pp. [3] | new pp. [3]

Before (including deletions): [16],

After: [18],

Change 100: replace | old pp. [3] | new pp. [3]

Before (including deletions): [17]

After: [19]

Change 101: replace | old pp. [3] | new pp. [3]

Before (including deletions): However, multiple independent benchmarks show

After: Connectivity-specific benchmarks report results

Change 102: replace | old pp. [3] | new pp. [3]

Before (including deletions): general VLMs handle component recognition far better than connectivity.

After: depend on the task, model and input conditions.

Change 103: replace | old pp. [3] | new pp. [3]

Before (including deletions): connection

After: connection

Change 104: replace | old pp. [3] | new pp. [3]

Before (including deletions): [15],

After: [17],

Change 105: replace | old pp. [3] | new pp. [3]

Before (including deletions): Gemini-2.5-Pro

After: Gemini-2.5-Pro

Change 106: replace | old pp. [3] | new pp. [3]

Before (including deletions): [14]

After: [16]

Change 107: replace | old pp. [3] | new pp. [3]

Before (including deletions): This corroborates the design choice, shared with SINA [3],

After: These task-specific results motivate a separately scoped connectivity diagnostic; they do not establish that geometry is superior

Change 108: replace | old pp. [3] | new pp. [3]

Before (including deletions): keep wire-to-component connectivity out of the VLM and instead solve it with an explicit geometric model. We make this separation concrete and quantify it with

After: VLMs on

Change 109: replace | old pp. [3] | new pp. [3]

Before (including deletions): own

After: corpus. Our

Change 110: replace | old pp. [3] | new pp. [3]

Before (including deletions): connectivity experiment.

After: experiment uses matched annotated component priors, as described in Section IV-G.

Change 111: replace | old pp. [3] | new pp. [3]

Before (including deletions): [18],

After: [20],

Change 112: replace | old pp. [3] | new pp. [3]

Before (including deletions): [19]

After: [21]

Change 113: replace | old pp. [3] | new pp. [3]

Before (including deletions): [20]. The dom- inant

After: [22]. Our legacy

Change 114: replace | old pp. [3] | new pp. [3]

Before (including deletions): in existing pipelines

After: baseline

Change 115: replace | old pp. [3] | new pp. [3]

Before (including deletions): union- find [21].

After: union-find [23].

Change 116: replace | old pp. [3] | new pp. [3]

Before (including deletions): over-merges

After: can over- merge

Change 117: replace | old pp. [3] | new pp. [3]

Before (including deletions): shorting

After: assigning both pins of a

Change 118: replace | old pp. [3] | new pp. [3]

Before (including deletions): parts

After: part

Change 119: replace | old pp. [3] | new pp. [3]

Before (including deletions): onto

After: to

Change 120: replace | old pp. [3] | new pp. [3]

Before (including deletions): preference,

After: pref- erence,

Change 121: replace | old pp. [3] | new pp. [3]

Before (including deletions): degree- budget

After: degree-budget

Change 122: replace | old pp. [3] | new pp. [3]

Before (including deletions): [22], [23]

After: [24], [25]

Change 123: replace | old pp. [3] | new pp. [3]

Before (including deletions): endpoints

After: end- points

Change 124: replace | old pp. [3] | new pp. [3]

Before (including deletions): neural

After: neu- ral

Change 125: replace | old pp. [3] | new pp. [3]

Before (including deletions): [24],

After: [26],

Change 126: replace | old pp. [3] | new pp. [3]

Before (including deletions): com- ponent

After: component

Change 127: replace | old pp. [3] | new pp. [3]

Before (including deletions): rule- based connectivity

After: rule-based connec- tivity

Change 128: replace | old pp. [3] | new pp. [3]

Before (including deletions): [25] pre- dicts

After: [27] predicts

Change 129: replace | old pp. [3] | new pp. [3]

Before (including deletions): are powerful but require

After: use

Change 130: replace | old pp. [3] | new pp. [3]

Before (including deletions): connec- tivity

After: connectivity

Change 131: replace | old pp. [3] | new pp. [3]

Before (including deletions): data and, trained on printed schematics, do

After: data. Transfer of the cited printed-schematic model to this hand-drawn benchmark is

Change 132: replace | old pp. [3] | new pp. [3]

Before (including deletions): transfer to hand-drawn images without retraining;

After: established here;

Change 133: replace | old pp. [3] | new pp. [3]

Before (including deletions): model needs

After: join requires

Change 134: replace | old pp. [3] | new pp. [3]

Before (including deletions): training and is the focus here.

After: training.

Change 135: replace | old pp. [3] | new pp. [3]

Before (including deletions): [26]

After: [28]

Change 136: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: struc- tural

Change 137: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: inferred

Change 138: replace | old pp. [3] | new pp. [3]

Before (including deletions): nodes, suitable for

After: nodes. Component-pair evaluation does not certify those pin assignments or the exact net partition.

Change 139: replace | old pp. [3] | new pp. [3]

Before (including deletions): netlist generation.

After: export is illustrative: component values and device models require external specification, and simulation equivalence on real scans is not validated.

Change 140: insert | old pp. [] | new pp. [3]

Before (including deletions): (none)

After: VOLUME 11, 2023 3

Change 141: replace | old pp. [3] | new pp. [4]

Before (including deletions): with weight 0. This ensures

After: so

Change 142: replace | old pp. [3] | new pp. [4]

Before (including deletions): of a wire are always in

After: belong to

Change 143: replace | old pp. [3] | new pp. [4]

Before (including deletions): VOLUME 11, 2023 3

After: The implementation accumulates connectivity with union-find; it does not solve a weighted graph optimization.

Change 144: replace | old pp. [4] | new pp. [4]

Before (including deletions): endpoint pairs (ei, ej)

After: endpoints ei, ej on different wires

Change 145: replace | old pp. [4] | new pp. [4]

Before (including deletions): edge with weight $\|e_i - e_j\|$.

After: an edge.

Change 146: replace | old pp. [4] | new pp. [4]

Before (including deletions): captures wire

After: proximity rule can connect

Change 147: replace | old pp. [4] | new pp. [4]

Before (including deletions): corners,

After: corners

Change 148: replace | old pp. [4] | new pp. [4]

Before (including deletions): junctions.

After: junctions, but can also merge unrelated nearby wires.

Change 149: replace | old pp. [4] | new pp. [4]

Before (including deletions): For each

After: Each

Change 150: replace | old pp. [4] | new pp. [4]

Before (including deletions): e

After: is connected to at most one selected pin by this edge rule. First apply the component-first assignment described below. Only if no component is assigned, search pins within Euclidean distance $d \leq \tau_{\text{pin}}$

Change 151: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: select the

Change 152: delete | old pp. [4] | new pp. []

Before (including deletions): p

After: (none)

Change 153: replace | old pp. [4] | new pp. [4]

Before (including deletions): $\|e-p\| \leq \tau_{\text{pin}}$, add edge

After: the smallest score. The score is d unless directional preference is enabled and $d > 10^{-6}$ px; in that case it is $d[1 - \alpha \max(0, \cos \theta)]$,

Change 154: replace | old pp. [4] | new pp. [4]

Before (including deletions): weight $\|e-p\| \cdot (1 - \alpha \cdot \cos \theta)$, where

After: fixed $\alpha = 0.35$ and

Change 155: replace | old pp. [4] | new pp. [4]

Before (including deletions): is

After: measured from

Change 156: replace | old pp. [4] | new pp. [4]

Before (including deletions): angle between the

After: outward

Change 157: replace | old pp. [4] | new pp. [4]

Before (including deletions): and

After: to

Change 158: replace | old pp. [4] | new pp. [4]

Before (including deletions): vector, and α controls directional preference. This

After: vector. Directional preference

Change 159: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: enabled in

Change 160: replace | old pp. [4] | new pp. [4]

Before (including deletions): critical

After: scale-relative base, but applies only to this fallback search. If an assigned component has no eligible pin, or the fallback finds none, no endpoint–pin

Change 161: replace | old pp. [4] | new pp. [4]

Before (including deletions): type:

After: is added. The score selects a fallback pin;

Change 162: replace | old pp. [4] | new pp. [4]

Before (including deletions): connects wires to components.

After: is not a weight optimized over the final graph.

Change 163: replace | old pp. [4] | new pp. [4]

Before (including deletions): each

After: an

Change 164: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: a different

Change 165: replace | old pp. [4] | new pp. [4]

Before (including deletions): $w_j = (c, d)$,

After: w_j , add an edge

Change 166: replace | old pp. [4] | new pp. [4]

Before (including deletions): point-to-segment

After: point-to- segment

Change 167: replace | old pp. [4] | new pp. [4]

Before (including deletions): $\text{dist}(e, w_j) \leq \tau$, add edge with weight $\text{dist}(e, w_j)$.

After: is at most τ and the normalized projection parameter lies in $[0.15, 0.85]$.

Change 168: replace | old pp. [4] | new pp. [4]

Before (including deletions): captures T-junctions where a wire terminates on another wire's body.

After: rule represents T-junction candidates away from the segment ends.

Change 169: replace | old pp. [4] | new pp. [4]

Before (including deletions): each

After: a

Change 170: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: add an edge

Change 171: replace | old pp. [4] | new pp. [4]

Before (including deletions): $\text{dist}(p, w_j) \leq \tau$

After: is at most τ

Change 172: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: normalized

Change 173: replace | old pp. [4] | new pp. [4]

Before (including deletions): lands mid-span (not at w_j 's endpoints, which are handled by edge type 3), add edge with weight $\text{dist}(p, w_j)$.

After: parameter lies in $[0.05, 0.95]$.

Change 174: replace | old pp. [4] | new pp. [4]

Before (including deletions): binds components that tap onto

After: rule represents pin-to-rail candidates along

Change 175: delete | old pp. [4] | new pp. []

Before (including deletions): passing rail along the

After: (none)

Change 176: replace | old pp. [4] | new pp. [4]

Before (including deletions): body rather than at its end. It recovers connectivity that endpoint-only binding misses, without the all-to-all over-merge of radius grabbing.

After: body; geometric proximity alone does not certify an electrical connection.

Change 177: replace | old pp. [4] | new pp. [4]

Before (including deletions): τ_{join} , τ_{pin} , and

After: are manually configured scale-relative rules with fixed pixel clamps (in pixels): $\tau_{\text{pin}} = \min(60, \max(24, 0.62s))$, $\tau_{\text{join}} = \min(28, \max(11, 0.30s))$,

Change 178: delete | old pp. [4] | new pp. []

Before (including deletions): are scale-relative: τ

After: (none)

Change 179: replace | old pp. [4] | new pp. [4]

Before (including deletions): $k \cdot s$ where

After: $\min(20, \max(8, 0.20s))$. Here

Change 180: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: component-bounding-box diagonal scale; the implementation uses the middle sorted diagonal (the upper middle for an even count), falling back to a wire-length

Change 181: replace | old pp. [4] | new pp. [4]

Before (including deletions): (median

After: when

Change 182: replace | old pp. [4] | new pp. [4]

Before (including deletions): diagonal), so

After: geometry is unavailable or degenerate. The multipliers and clamps are fixed, not learned. Because s is

Change 183: replace | old pp. [4] | new pp. [4]

Before (including deletions): parameter setting works across circuits of varying size.

After: scalar per image, intra-image size variance is not modeled: a schematic mixing a large IC with small discrete symbols shares one tolerance per rule. Extreme-size performance was not evaluated by controlled image rescaling.

Change 184: replace | old pp. [4] | new pp. [4]

Before (including deletions): assign

After: compute the distance from

Change 185: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: each component's oriented bounding box (OBB): 0 inside the rectangle, otherwise the minimum distance to

Change 186: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: edges. If four OBB vertices are unavailable, use axis-aligned bounding-box (AABB) distance instead. Assign the endpoint to the

Change 187: replace | old pp. [4] | new pp. [4]

Before (including deletions): by bounding-box

After: whose

Change 188: replace | old pp. [4] | new pp. [4]

Before (including deletions): (0 if inside the bbox, using a generous radius of

After: is within

Change 189: replace | old pp. [4] | new pp. [4]

Before (including deletions): diagonal)).

After: diagonal), where the diagonal is computed from that component's AABB.

Change 190: replace | old pp. [4] | new pp. [4]

Before (including deletions): by geometric position: for horizontal components ($w > h$), left pin if endpoint $x < \text{center } x$, else right pin; for vertical components ($h > w$), top pin if endpoint $y < \text{center } y$, else bottom pin. This handles

After: with

Change 191: replace | old pp. [4] | new pp. [4]

Before (including deletions): common case where endpoints land inside a component's bounding box but far from its pin.

After: smallest Euclidean distance to the endpoint.

Change 192: replace | old pp. [4] | new pp. [4]

Before (including deletions): endpoint-graph

After: endpoint-graph

Change 193: delete | old pp. [4] | new pp. []

Before (including deletions): some component pins "floating": touching only their own component's net, with no wire connecting them to another component. This indicates

After: (none)

Change 194: delete | old pp. [4] | new pp. []

Before (including deletions): dropped or over-displaced wire endpoint that the detector failed to place near the correct pin. Degree-budget completion identifies these floating pins and reconnects them via constrained optimization. A

After: (none)

Change 195: replace | old pp. [4] | new pp. [4]

Before (including deletions): p is treated as floating if its

After: in a

Change 196: replace | old pp. [4] | new pp. [4]

Before (including deletions): contains

After: containing

Change 197: replace | old pp. [4] | new pp. [4]

Before (including deletions): p's

After: its

Change 198: replace | old pp. [4] | new pp. [4]

Before (including deletions): component,

After: component. Completion treats such pins as candidates for recovery,

Change 199: replace | old pp. [4] | new pp. [4]

Before (including deletions): gnd, vss,

After: ground, supply markers,

Change 200: replace | old pp. [4] | new pp. [4]

Before (including deletions): junction,

After: probes, junctions

Change 201: replace | old pp. [4] | new pp. [4]

Before (including deletions): terminal. For

After: terminals. This condition can reflect a missing connection, but does not establish that the detector dropped a wire. Degree-budget completion retains the base net assignments and constructs candidates from

Change 202: replace | old pp. [4] | new pp. [4]

Before (including deletions): p, we generate candidate edges

After: p

Change 203: delete | old pp. [4] | new pp. []

Before (including deletions): all

After: (none)

Change 204: replace | old pp. [4] | new pp. [4]

Before (including deletions): in

After: on

Change 205: replace | old pp. [4] | new pp. [4]

Before (including deletions): within reach

After: and different current nets. Candidate distance must not exceed

Change 206: replace | old pp. [4] | new pp. [4]

Before (including deletions): $p \cdot s$,

After: $p\tau c$,

Change 207: replace | old pp. [4] | new pp. [4]

Before (including deletions): p is the reach factor; we use p

After: τc

Change 208: replace | old pp. [4] | new pp. [4]

Before (including deletions): 4 for

After: $\min(60, \max(24, 0.62s))$ px and

Change 209: replace | old pp. [4] | new pp. [4]

Before (including deletions): join,

After: completion uses $p = 4$. Thus reach uses the clamped pin scale, not the raw component diagonal; completion retains this scale dependence when base tolerances are ablated. For each candidate, a wire witness is sought by pairing the end- points of a detected segment with p, q in either orientation. The witness cost is the smallest, over feasible segments

Change 210: replace | old pp. [4] | new pp. [4]

Before (including deletions): Section IV-D shows

After: orientations, of the larger endpoint-to-pin distance, bounded by r . If no witness exists, the deployed relaxed-witness mode uses $\|p - q\|$ as the candidate cost. When candidates exist, the implementation builds

Change 211: replace | old pp. [4] | new pp. [4]

Before (including deletions): broad F1 plateau for $p \in [3, 5]$. Candidate edge weight is Euclidean distance $\|p - q\|$. We then solve a min-cost b-matching problem, $\min_{(p,q) \in E'} w(p, q) \cdot x_{pq}$ subject to $\sum q x_{pq} \leq 1$ for each

After: rectangular assignment matrix with one row per

Change 212: replace | old pp. [4] | new pp. [4]

Before (including deletions): pin p , using the Hungarian algorithm [22] (`scipy.optimize.linear_sum_assignment`).

After: pin.

Change 213: replace | old pp. [4] | new pp. [4]

Before (including deletions): offers

After: supplies

Change 214: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: distinct slot columns, limited by its candidate demand, and each row has a private dummy column with cost $1.5r$ for opting out. Unavailable assignments carry sentinel cost 106. The Hungarian algorithm [24] (`scipy.optimize.linear_sum_assignment`) assigns every row to exactly one column, with at most one row per column. This mandatory assignment with dummy options avoids the empty-solution ambiguity of a nonnegative-cost objective with only at-most-one constraints; target

Change 215: replace | old pp. [4] | new pp. [4]

Before (including deletions): so legitimate multi-terminal

After: implement the bounded-capacity matching [25]. Matches are then applied sequentially. Dummy or unavailable assignments add no edge. A proposed merge is also skipped if its endpoints are already in the same net or if their current

Change 216: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: share a component. This guard limits additional completion merges; it does not undo incorrect base unions or guarantee globally short-free output. Rejected matches

Change 217: replace | old pp. [4] | new pp. [4]

Before (including deletions): starved, and a per-pin opt-out leaves a pin with no good target unconnected rather than forcing a spurious edge. Finally, we reject any matched edge (p, q) if p and q belong

After: reassigned, so the final guarded result is not claimed

Change 218: insert | old pp. [] | new pp. [4]

Before (including deletions): (none)

After: minimize

Change 219: replace | old pp. [4] | new pp. [4]

Before (including deletions): same component or if adding the edge would merge two nets that already share a component, which structurally prevents short-circuiting two-terminal parts.

After: assignment objective over all valid netlists.

Change 220: insert | old pp. [] | new pp. [4, 5]

Before (including deletions): (none)

After: 4 VOLUME 11, 2023 (a) Radius-based joining R1 C1 R2 radius grabs every pin in range →R1, C1, R2 wrongly merged (b) Endpoint-graph (ours) R1 C1 R2 Wire body Endpoint–pin Endpoint–endpoint T-junction (endpt–body) Rail-tap (pin–body) Endpoint Pin FIGURE 3. Joining approaches compared. (a) Radius-based join (legacy): each wire endpoint grabs all pins within a fixed threshold, causing over-merging. (b) Our endpoint-graph model: typed edges and component-first pin assignment generate connectivity candidates, with directional scoring confined to the fallback pin search. Incorrect base merges remain possible.

Change 221: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: unordered

Change 222: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: This metric discards terminal identity: even a perfect component-pair score does not certify pin assignment, the exact net partition, absence of shorts, or simulation equivalence.

Change 223: replace | old pp. [4] | new pp. [5]

Before (including deletions): simulation

After: simulation

Change 224: replace | old pp. [4] | new pp. [5]

Before (including deletions): pipeline

After: historical 16 px-anchor configuration

Change 225: replace | old pp. [4] | new pp. [5]

Before (including deletions): = 1.000 across the benchmark set,

After: 1.000,

Change 226: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: benchmark

Change 227: replace | old pp. [4] | new pp. [5]

Before (including deletions): ≥ 0.90 . Sauvola adaptive binarization dominates all alternatives:

After: at least 0.90. Its aggregate F1 is 0.9755 at 10■/18 px overlap deduplication; the 12■/8 px production configuration gives 0.9726 with the same 16 px anchor distance. Among the tested alternatives,

Change 228: replace | old pp. [4] | new pp. [5]

Before (including deletions): = 0.789,

After: 0.789 and

Change 229: delete | old pp. [4] | new pp. []

Before (including deletions): F1 =

After: (none)

Change 230: delete | old pp. [4, 5] | new pp. []

Before (including deletions): Component occlusion matters: without it, component internals (text, symbols, internal wiring) generate hundreds of false wire detections. 4 VOLUME 11, 2023 (a) Radius-based joining R1 C1 R2 radius grabs every pin in range →R1, C1, R2 wrongly merged (b) Endpoint-graph (ours) R1 C1 R2 Wire body Endpoint–pin Endpoint–endpoint T-junction (endpt–body) Rail-tap (pin–body) Endpoint Pin FIGURE 3. Joining approaches compared. (a) Radius-based join (legacy): each wire endpoint grabs all pins within a fixed threshold, causing over-merging. (b) Our endpoint-graph model: typed edges (wire body, endpoint–pin) with directional preference resolve connectivity without over-merging. Before completion R1 R2 R3 Net A floating pin After completion R1 R2 R3 Net A completion edge (min-cost b-matching) self-loop guard: a component's own two terminals are never matched together FIGURE 4. Degree-budget completion. Left: After base joining, some pins are “floating” (touching only their own component). Right: Completion reconnects floating pins via min-cost b-matching (orange dashed), with self-loop guard preventing shorts within two-terminal components. 0.6 0.7 0.8 0.9 1.0 Wire detection F1 (134 images) Triangle Otsu adaptive Gaussian Sauvola (alt.) Sauvola (k=0.2875) Sauvola (w=61) Sauvola, anchor 12px Sauvola, anchor 16px (ours) 0.758 0.789 0.928 0.949 0.950 0.959 0.973 0.976 FIGURE 5. Wire detection F1 across thresholding methods and pipeline variants (134 images). Our configuration (Sauvola binarization with a 16 px anchor filter) achieves F1 = 0.976. Otsu and triangle thresholding degrade significantly.

After: (none)

Change 231: replace | old pp. [5] | new pp. [5]

Before (including deletions): Detection Performance (134 images)

After: detection performance on 134 images with annotated component boxes. The two Sauvola anchor-distance rows use historical 10■/18 px overlap deduplication; they are not the production 12■/8 px configuration.

Change 232: insert | old pp. [] | new pp. [5]

Before (including deletions): (none)

After: Occlusion is intended to remove component internals before wire extraction, but a full occlusion-removal ablation is not reported here; no quantified gain from that intervention is claimed.

Change 233: delete | old pp. [5] | new pp. []

Before (including deletions): The completion-based strategies are the most robust, maintaining F1 ≥ 0.94 at the highest error severity (L4), compared to F1 = 0.36 for the legacy radius union-find baseline, a 2.6×

After: (none)

Change 234: replace | old pp. [5] | new pp. [5]

Before (including deletions): improvement. The radius join's transitive grabbing causes catastrophic over-merging under noise:

After: On the authored synthetic suite

Change 235: replace | old pp. [5] | new pp. [5, 6]

Before (including deletions): it

After: the scale-relative graph with completion scores F1 0.95, rescue graph with completion 0.94, rescue graph alone 0.90, scale-relative graph alone 0.85 and radius union-find 0.36 (Table 2). These results compare the reported configurations under the specified perturbations. They do not establish that completion repairs every dropped connection or prevents all shorts. The scale-relative completion configuration also has the highest component-pair score among the deterministic configurations reported on the real-image benchmark (Section IV-D). For readability, two intermediate synthetic variants whose scores

	R1	R2	R3	Net A
Before completion	0.6	0.7	0.8	0.9
Completion edge (min-cost b-matching)	0.845	0.949	0.950	0.959
self-loop guard: a component's own two terminals are never matched together	0.973	0.976		

FIGURE 4. Degree-budget completion. Left: After base joining, some pins are “floating” (touching only their own component). Right: Completion reconnects floating pins via min-cost b-matching (orange dashed), with a guard rejecting additional merges between nets that already share a component; existing base

Change 236: replace | old pp. [5] | new pp. [6]

Before (including deletions): multiple components onto shared nets. We emphasize, however, that

After: are not repaired. 0.6 0.7 0.8 0.9 1.0 Wire detection F1 (134 images) Triangle (skeleton) Otsu adaptive Gaussian (skeleton) Sauvola (alt.) Sauvola (k=0.2875) Sauvola (w=61) Sauvola, anchor 12px Sauvola, anchor 16px (ours) 0.758 0.789 0.845 0.949 0.950 0.959 0.973 0.976 FIGURE 5. Wire detection F1 across

Change 237: replace | old pp. [5] | new pp. [6]

Before (including deletions): endpoint-graph base joins are themselves far more robust than the radius join, reaching

After: tested thresholding methods and pipeline variants (134 images, annotated component boxes). The historical Sauvola configuration with a 16 px anchor filter and overlap deduplication at 10/18 px achieves

Change 238: replace | old pp. [5] | new pp. [6]

Before (including deletions): 0.90 at L4 (Table 2).

After: 0.976; this is distinct from the production 12/8 px deduplication setting.

Change 239: replace | old pp. [5] | new pp. [6]

Before (including deletions): degree-budget completion adds a further +0.05 over

After: adaptive Gaussian and Triangle bars use

Change 240: replace | old pp. [5] | new pp. [6]

Before (including deletions): strongest of these by recovering dropped connections the base graph join leaves floating, while its per-pin edge budget keeps precision high. Pairing that completion with the high-precision scale-relative base (the scale-relative completion join used throughout) gives the best synthetic robustness (0.95 at L4) and, more importantly, the best real-image connectivity (Section IV-D). The large gap is against the legacy radius join; against the strongest prior graph join the improvement is smaller but consistent across every severity level. For readability we omit two intermediate variants whose synthetic scores fall within rounding of

After: stored skeleton-extraction configurations (F1 = 0.845 and 0.758, respectively). round to

Change 241: replace | old pp. [5] | new pp. [6]

Before (including deletions): (nearest-pin without anchoring,

After: are omitted: nearest-pin without anchoring,

Change 242: replace | old pp. [5] | new pp. [6]

Before (including deletions): rescue).

After: rescue.

Change 243: replace | old pp. [5] | new pp. [6]

Before (including deletions): a parametric

After: an authored

Change 244: replace | old pp. [5] | new pp. [6]

Before (including deletions): To validate

After: For the real-image evaluation, we constructed a human-verified net-level benchmark of 31 CGHD- 1152 images using a purpose-built verification tool. Thirty-four images were annotated; three were excluded before scoring (one pending human verification, two failing pre-registered physical-sanity checks), leaving the scored $N=31$. Net-level proposals were bootstrapped by a conservative endpoint-graph join over the dataset's annotated wires and then corrected by one human annotator. Human correction and evaluation

Change 245: insert | old pp. [] | new pp. [6]

Before (including deletions): (none)

After: independently authored synthetic ground truth mitigate concern about method-dependent labels, but do not rule out residual bootstrap or annotator bias in the

Change 246: replace | old pp. [5] | new pp. [6]

Before (including deletions): detector output, we constructed,

After: benchmark. Both the geometric join and the VLM diagnostic receive annotated component boxes. Connectivity is scored as component-pair F1 restricted

Change 247: delete | old pp. [5] | new pp. []

Before (including deletions): our knowledge,

After: (none)

Change 248: replace | old pp. [5] | new pp. [6]

Before (including deletions): first

After: evaluated electrical- component subset, discarding terminal identities as in the synthetic connectivity metric. Table 4 reports the evaluated join configurations on the verified set, run on the pipeline's own detected wires. We report micro-averaged (pair-pooled) component-pair F1 as the primary metric, so that a dense 14-component board contributes in proportion to its connections rather than counting the same as a 3-component one; the macro (mean per-image) F1 is listed alongside for reference. The best configuration, the scale-relative completion join (degree-budget completion applied to a high-precision scale-relative endpoint-graph base), achieves micro-F1 = 0.890 ($P = 0.92$, $R = 0.86$; 95% bootstrap CI [0.855, 0.924]), improving on the prior default that applies the same completion to the rescue base (0.829) and the radius union-find join (0.667) (Figure 7). The scale- relative completion configuration omits the rescue base's 12 px end-extension and dead-end rescue and has higher reported precision. This comparison does not isolate the causal contribution of those individual changes. A bounded within- benchmark reach sweep over $p \in [3, 5]$, with $r = p\tau c$, yields macro-F1 0.895–0.903. This is sensitivity evidence for the tested settings, not a controlled image-rescaling or mixed- size robustness experiment. Confidence intervals are wide at $N = 31$ and the per-strategy intervals overlap, but the ranking is consistent and is corroborated on the independently authored synthetic suite (Table 2). Table 3 reports bounded interventions on the base graph and the resulting pipeline with completion. Disabling T- junctions, rail taps or directional preference leaves the displayed base and full aggregate counts unchanged. Fixed-pixel base tolerances increase base micro-F1 from 0.816 to 0.820; the full-pipeline counts again tie at 418/37/66. Equal aggregate counts do not establish identical recovered connections, explain which mechanisms were active, or prove that completion masks differences. The baseline comparison shows an increase from base micro-F1 0.816 to 0.890 with completion, but these ablations do not establish that other mechanisms are generally unnecessary. Full occlusion removal and removal of scale dependence throughout completion were not tested. The ablation evidence therefore does not cover every proposed module or extreme-size input condition. This real-image ranking resolves an open question from prior work on this pipeline: against bootstrapped pilot labels the degree-budget completion had not consistently beaten its base join; against

Change 249: replace | old pp. [5] | new pp. [6]

Before (including deletions): net-

After: ground truth it clearly does (micro-F1 0.829 vs 0.787 for the rescue base alone), and the scale-relative variant is better still. The same ordering holds on the authored synthetic suite (Table 2), where the scale- relative completion join also ranks first, providing corroborat-

Change 250: delete | old pp. [5] | new pp. []

Before (including deletions): 5

After: (none)

Change 251: replace | old pp. [6] | new pp. [7]

Before (including deletions): Errors cause fragmentation rather than shorts (precision stays high; Table 4);

After: High precision does not exclude false merges or certify pin correctness;

Change 252: delete | old pp. [6] | new pp. []

Before (including deletions): balanced

After: (none)

Change 253: replace | old pp. [6] | new pp. [7]

Before (including deletions): is

After: has

Change 254: replace | old pp. [6] | new pp. [7]

Before (including deletions): hardest case.

After: lowest reported connectivity F1 and simulation accuracy (Table 6). TABLE 3. Leave-one-out ablation of endpoint-graph rules at the base-graph

Change 255: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: and with completion, using annotated component boxes and detected wires on the 31-image benchmark. All full-pipeline rows have the same aggregate TP/FP/FN (418/37/66). Fixed-pixel base tolerances give micro-F1 0.820 versus 0.816 for the scale-relative base; completion retains its scale dependence. Aggregate ties do not establish identical recovered connections or a causal explanation for the ties. Configuration Base graph only Full pipeline micro-F1 TP/FP/FN micro-F1 P R Baseline (all edge types, scale-relative) 0.816 352/27/132 0.890 0.919 0.864 T-junctions disabled 0.816 352/27/132 0.890 0.919 0.864 Rail taps disabled 0.816 352/27/132 0.890 0.919 0.864 T-junctions + rail taps disabled 0.816 352/27/132 0.890 0.919 0.864 Directional preference disabled 0.816 352/27/132 0.890 0.919 0.864 Fixed-pixel base tolerances; completion unchanged 0.820 356/28/128 0.890 0.919 0.864 TABLE 4. Join strategy and baseline comparison on human-verified net-level

Change 256: delete | old pp. [6] | new pp. []

Before (including deletions): for hand-drawn circuits: 31 images from CGHD-1152 in which every electrical net was traced and confirmed by a human annotator through a purpose-built verification tool. Because one of our comparison points is a vision-language model (Section IV-F), the human, not any model, is the verifier of record, so the answer key is independent of every method under test. Net-level proposals were bootstrapped by a conservative endpoint-graph join over the dataset's perfect wire labels and then corrected by hand; during verification the human consistently caught cases where the bootstrap join had wrongly split a single rail, and we additionally re-validate all conclusions on the independently authored synthetic ground truth to rule out residual bootstrap bias. Connectivity is scored as component-pair F1 restricted to SPICE-active components, identical to the synthetic metric. Table 3 reports every registered join strategy on the verified set, run on the pipeline's own detected wires. We report micro-averaged (pair-pooled) component-pair F1 as the primary metric, so that a dense 14-component board contributes in proportion to its connections rather than counting the same as a 3-component one; the macro (mean per-image) F1 is listed alongside for reference. The best configuration, the scale-relative completion join (degree-budget completion applied to a high-precision scale-relative endpoint-graph base), achieves micro-F1 = 0.890 (P = 0.92, R = 0.86; 95% bootstrap CI [0.855, 0.924]), improving on the prior default that applies the same completion to the rescue base (0.829) and the radius union-find join (0.667) (Figure 7). The gain over that rescue-based completion comes from the cleaner base: the rescue base's 12 px end-extension and dead-end rescue over-reach on noisy detected wires and cost precision, whereas the scale-relative base keeps precision high while the bounded completion still recovers recall. A reach sweep shows a broad F1 plateau across reach 3–5×s, so the setting is robust rather than tuned to a spike. Confidence intervals are wide at N = 31 and the per-strategy intervals overlap, but the ranking is consistent and is corroborated on the independently authored synthetic suite (Table 2). TABLE 3. Join Strategy and Baseline Comparison on Human-Verified Net-Level Ground Truth

After: (none)

Change 257: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: annotated component boxes;

Change 258: replace | old pp. [6] | new pp. [7]

Before (including deletions): wires).

After: wires except for the stated perfect-wire reference).

Change 259: replace | old pp. [6] | new pp. [7]

Before (including deletions): Reference (non-deterministic

After: Conditional VLM diagnostic

Change 260: replace | old pp. [6] | new pp. [7]

Before (including deletions): oracle wires)

After: annotated-wire reference

Change 261: replace | old pp. [6] | new pp. [7]

Before (including deletions): This real-image ranking resolves an open question

After: ing evidence on independently authored data without ruling out residual bias in the real benchmark labels. Several image-to-netlist systems erase component regions and infer nets

Change 262: replace | old pp. [6] | new pp. [7]

Before (including deletions): prior work

After: the remaining wire pixels using connected components or contours, including SINA [3] and Peker et al. [5]. We evaluate a connected-component baseline

Change 263: replace | old pp. [6] | new pp. [7]

Before (including deletions): this pipeline: against bootstrapped pilot labels

After: our detected wire segments to control

Change 264: replace | old pp. [6] | new pp. [7]

Before (including deletions): degree-budget completion had not consistently beaten its base join; against human-verified ground truth it clearly does (micro-F1 0.829 vs 0.787

After: wire input. Its best tested configuration reaches component-pair micro-F1 0.624, versus 0.890

Change 265: replace | old pp. [6] | new pp. [7]

Before (including deletions): rescue base alone), and the scale-relative variant is better still. The same ordering holds

After: endpoint-graph join (Table 4). Hough line fitting [20] followed by proximity linking reaches 0.805 at its best tested linking tolerance

Change 266: replace | old pp. [6] | new pp. [7]

Before (including deletions): authored synthetic suite (Table 2), where

After: same detected wires. These are

Change 267: replace | old pp. [6] | new pp. [7]

Before (including deletions): scale- relative completion join also ranks first, so

After: best configurations in

Change 268: replace | old pp. [6] | new pp. [7]

Before (including deletions): result is

After: reported sweeps,

Change 269: replace | old pp. [6] | new pp. [7]

Before (including deletions): an artifact of how the net-level labels were bootstrapped. The dominant connectivity recipe in prior image-to-netlist systems erases component boxes and reads nets off the connected components, or contours,

After: upper bounds on all implementations

Change 270: replace | old pp. [6] | new pp. [7]

Before (including deletions): residual wire pixels. It

After: classical methods. Their ordering

Change 271: replace | old pp. [6] | new pp. [7]

Before (including deletions): used by SINA [3],

After: evidence for these configurations and this benchmark, not a direct performance comparison with

Change 272: replace | old pp. [6] | new pp. [7]

Before (including deletions): concurrent contour-based 6

After: cited complete systems. We additionally attempted to run five public image-to-netlist systems on the hand-drawn benchmark and encountered the following implementation, model-vocabulary and input-domain limitations. The classical printed-schematic interpreter of Kelly and Cole [2] installs and runs, but on hand-drawn scans it over-segments wires into spurious components and recovers almost no connectivity, a domain-transfer failure. Netlistify [27] reproduces on its own printed AMS test data, which we ran end-to-end with the released model, but its active model lacks the diode, inductor, and IC classes our circuits contain and its connectivity model is trained on printed thin-line appearance, limiting its applicability to the device population in this benchmark. Masala-CHAI [4] del-

Change 273: insert | old pp. [] | new pp. [7]

Before (including deletions): (none)

After: 7

Change 274: replace | old pp. [7] | new pp. [8]

Before (including deletions): Our

After: The

Change 275: replace | old pp. [7] | new pp. [8]

Before (including deletions): leads all

After: has the highest score among the evaluated

Change 276: replace | old pp. [7] | new pp. [8]

Before (including deletions): strategies

After: configurations

Change 277: delete | old pp. [7] | new pp. []

Before (including deletions): the

After: (none)

Change 278: delete | old pp. [7] | new pp. []

Before (including deletions): Hough and connected-component

After: (none)

Change 279: delete | old pp. [7] | new pp. []

Before (including deletions): (run

After: (none)

Change 280: replace | old pp. [7] | new pp. [8]

Before (including deletions): identical

After: these

Change 281: replace | old pp. [7] | new pp. [8]

Before (including deletions): wires).

After: wires.

Change 282: replace | old pp. [7] | new pp. [8]

Before (including deletions): a strong

After: the

Change 283: insert | old pp. [] | new pp. [8]

Before (including deletions): (none)

After: diagnostic

Change 284: replace | old pp. [7] | new pp. [8]

Before (including deletions): 4.8) evaluated on

After: 4.8). Both methods receive annotated component boxes;

Change 285: replace | old pp. [7] | new pp. [8]

Before (including deletions): same 31 images, from which our join

After: VLM-minus-join difference

Change 286: replace | old pp. [7] | new pp. [8]

Before (including deletions): statistically indistinguishable (paired micro-F1 difference +0.033, 95%

After: +0.033 (95%)

Change 287: replace | old pp. [7] | new pp. [8]

Before (including deletions): +0.078]) while being deterministic, simulatable, and orders of magnitude cheaper.

After: +0.078]). The interval does not establish equivalence, cost superiority or a global validity guarantee.

Change 288: replace | old pp. [7] | new pp. [8]

Before (including deletions): node detector

After: 3 4 5 7 10 11 12 14 SPICE-active components per image 0 1 2 3 4 5 6 7 # images 15 images 5 comps 12 images 10 comps 3 6 6 4 6 1 3 2 e 31-image human-verified benchmark (median 7, max 14) FIGURE 8. Complexity

Change 289: replace | old pp. [7] | new pp. [8]

Before (including deletions): Peker et al. [5], AMSnet [13],

After: the 31-image human-verified benchmark, counting components in the evaluated electrical subset: 15 images have ≤ 5

Change 290: replace | old pp. [7] | new pp. [8]

Before (including deletions): the Bayer instance-segmentation pipeline [6], and was designed for clean printed schematics. On hand-drawn photographs it has no precise operating point: pencil-stroke fragmentation splits each net into many components ($F1 < 0.25$ at sensible morphological gap-bridging), and only pathological closing that effectively merges by proximity lifts recall, at a steep precision cost. For a controlled comparison that isolates the join algorithm from binarization

quality, we run connected- component labeling

After: 12 have ≥ 10 (median 7, range 3–14). Component-count breadth does not validate dense buses or multilayer crossings; performance

Change 291: delete | old pp. [7] | new pp. []

Before (including deletions): our own detected wire segments, identical input to our join, and its best configuration reaches only micro-F1 = 0.624 (recall collapses to 0.46: the blobs short or drop most nets), versus 0.890 for the endpoint-graph join (Table 3). A stronger classical alternative, Hough-transform line detection [18] followed by proximity linking, fares better, since line fitting denoises the fragmented strokes, and reaches micro-F1 = 0.805 at its best-tuned linking tolerance, but still trails our join by 0.085 at markedly lower precision. Each baseline is reported at its best configuration over a tolerance sweep, so

After: (none)

Change 292: replace | old pp. [7] | new pp. [8]

Before (including deletions): are upper bounds on the classical recipes. Explicit typed-endpoint reasoning with bounded completion thus outperforms both blob/contour connectivity and line- primitive proximity tracing. We additionally attempted to run five public image-to- netlist systems end-to-end on our hand-drawn benchmark, and none could be fairly applied. The classical printed- schematic interpreter of Kelly and Cole [2] installs and runs, but on hand-drawn scans it over-segments wires into spurious components and recovers almost no connectivity, a domain- transfer failure. Netlistify [25] reproduces on its own printed AMS test data, which we ran end-to-end with the released model, but its active model lacks the diode, inductor, and IC classes our circuits contain and its connectivity model is trained on printed thin-line appearance, so it cannot represent hand-drawn discrete circuits. Masala-CHAI [4] delegates

After: cases remains untested. egates

Change 293: replace | old pp. [7] | new pp. [8]

Before (including deletions): equivalent

After: equivalent

Change 294: replace | old pp. [7] | new pp. [8]

Before (including deletions): IV-F,

After: IV-G,

Change 295: replace | old pp. [7] | new pp. [8]

Before (including deletions): [26]

After: [28]

Change 296: replace | old pp. [7] | new pp. [8]

Before (including deletions): Co- lab

After: Colab

Change 297: replace | old pp. [7] | new pp. [8]

Before (including deletions): This survey is itself informative:

After: These observations concern the versions and configurations examined; they do not establish that

Change 298: replace | old pp. [7] | new pp. [8]

Before (including deletions): provides a deterministic, hand- drawn-ready connectivity module, so the controlled

After: can recover hand-drawn connectivity. The

Change 299: replace | old pp. [7] | new pp. [8]

Before (including deletions): baselines above, both run on our identical detected wires, are the fair points of comparison, and our join exceeds both. Running the best join

After: experiments above provide controlled comparisons

Change 300: replace | old pp. [7] | new pp. [8]

Before (including deletions): dataset's perfect wire labels rather than

After: same detected wires. With annotated component boxes held fixed, replacing

Change 301: replace | old pp. [7] | new pp. [8]

Before (including deletions): leaves

After: by the dataset's annotated wires changes component-pair

Change 302: replace | old pp. [7] | new pp. [8]

Before (including deletions): unchanged at 0.890,

After: from 0.8903 to 0.8898,

Change 303: delete | old pp. [7] | new pp. []

Before (including deletions): raising

After: (none)

Change 304: replace | old pp. [7] | new pp. [8]

Before (including deletions): only

After: increases

Change 305: replace | old pp. [7] | new pp. [8]

Before (including deletions): (Table 3), so the wire detector accounts for essentially zero pair-weighted F1.

After: (Table 4).

Change 306: replace | old pp. [7] | new pp. [8]

Before (including deletions): residual error is intrinsic join ambiguity on hand-drawn geometry, not wire detection, consistent with the 0.976 wire-detection F1 reported above (Table 1). The join degrades only gently with circuit size, with per-image F1 falling from 0.94 on circuits with ≤ 6 components to 0.86 above that and only weak correlation $r = -0.19$, and remains high-precision throughout, in contrast

After: micro scores round

Change 307: replace | old pp. [7] | new pp. [8]

Before (including deletions): VLM, whose precision collapses

After: same displayed value, but are not identical. This comparison concerns wire input conditional

Change 308: insert | old pp. [] | new pp. [8]

Before (including deletions): (none)

After: component annotations; it does not establish an autonomous detection bottleneck or exclude compensating changes in recovered connections. E. DESCRIPTIVE PER-DRAFTER RESULTS Table 5 reports

Change 309: replace | old pp. [7] | new pp. [8]

Before (including deletions): densest circuits (Section IV-F). E.

After: scale-relative completion join grouped by CGHD drafter. Drafter identities were recovered from the source tree for 16 of 31 images by direct filename match; the remaining 15 were inferred from the dataset's documented 12-circuits-per-drafter block structure. Four groups contain at least four images, with micro-F1 ranging from 0.854 to 0.987. These groups describe within-corpus variation, not evidence of unseen-drafter transfer. No held-out-drafter experiment was performed; the small, partly inferred groups do not support strong statistical separation. TABLE 5. Descriptive join micro-F1 (scale-relative completion) by CGHD drafter. Groups with at least four images; overall includes all benchmark images. Drafter identities include direct and inferred assignments. Drafter n images micro-F1 macro-F1 TP/FP/FN drafter_1 7 0.933 0.907 56/6/2 drafter_2 5 0.854 0.874 85/14/15 drafter_10 5 0.987 0.995 37/1/0 drafter_12 4 0.913 0.938 21/0/4 Overall 31 0.890 0.901

418/37/66 F.

Change 310: replace | old pp. [7] | new pp. [8]

Before (including deletions): 4

After: 6

Change 311: replace | old pp. [7] | new pp. [8]

Before (including deletions): 4.

After: 6.

Change 312: replace | old pp. [7] | new pp. [8]

Before (including deletions): remains high (≥ 0.85)

After: is at least 0.85

Change 313: insert | old pp. [] | new pp. [8]

Before (including deletions): (none)

After: the reported synthetic circuits at L4. This does not exclude false merges or establish that

Change 314: delete | old pp. [7] | new pp. []

Before (including deletions): circuits even at L4:

After: (none)

Change 315: replace | old pp. [7] | new pp. [8]

Before (including deletions): cause fragmentation (dropped connections) rather than shorts (false merges).

After: are fragmentation.

Change 316: replace | old pp. [7] | new pp. [8]

Before (including deletions): is

After: has

Change 317: replace | old pp. [7] | new pp. [8]

Before (including deletions): hardest topology ($F1 = 0.82$): its balanced arms make the inferred node structure sensitive to a single mis-placed connection,

After: lowest reported $F1$ (0.82)

Change 318: replace | old pp. [7] | new pp. [8]

Before (including deletions): falls to 6%. F.

After: of 6%, illustrating that component-pair connectivity and the separately evaluated synthetic simulation criterion measure different properties. G.

Change 319: replace | old pp. [7] | new pp. [8]

Before (including deletions): To quantify

After: We evaluate Claude Opus 4.8 as an oracle-component-box connectivity diagnostic. Both methods receive

Change 320: replace | old pp. [7] | new pp. [8]

Before (including deletions): separation argued in Section II, that connectivity is better solved by an explicit

After: same annotated component boxes; the

Change 321: replace | old pp. [7] | new pp. [8]

Before (including deletions): model than left to a VLM, we ran a first-party experiment with a strong general VLM (Claude Opus 4.8). To make the comparison fair, the

After: method uses detected wires. The

Change 322: delete | old pp. [7] | new pp. []

Before (including deletions): exactly what the join receives:

After: (none)

Change 323: delete | old pp. [7] | new pp. []

Before (including deletions): the same ground-truth

After: (none)

Change 324: delete | old pp. [7] | new pp. []

Before (including deletions): VOLUME 11, 2023 7

After: (none)

Change 325: replace | old pp. [8] | new pp. [8]

Before (including deletions): index,

After: index

Change 326: delete | old pp. [8] | new pp. []

Before (including deletions): is

After: (none)

Change 327: replace | old pp. [8] | new pp. [8]

Before (including deletions): return, as JSON, the

After: return JSON

Change 328: replace | old pp. [8] | new pp. [8]

Before (including deletions): that share

After: sharing

Change 329: replace | old pp. [8] | new pp. [8]

Before (including deletions): node; we

After: node. Each image uses a separate model call without shared conversational context. We

Change 330: replace | old pp. [8] | new pp. [8, 9]

Before (including deletions): component-pairs with

After: unordered component pairs 8 VOLUME 11, 2023 against

Change 331: replace | old pp. [8] | new pp. [9]

Before (including deletions): metric. Each image

After: human-verified benchmark. This

Change 332: replace | old pp. [8] | new pp. [9]

Before (including deletions): read by

After: a con- ditional connectivity comparison, not

Change 333: replace | old pp. [8] | new pp. [9]

Before (including deletions): independent, context-free model call (one per image), so no answer leaks between circuits, and

After: autonomous end-to-end comparison. On

Change 334: replace | old pp. [8] | new pp. [9]

Before (including deletions): human, not the model, remains the verifier of record. On a clean synthetic control (15 authored circuits)

After: 31 real images,

Change 335: delete | old pp. [8] | new pp. []

Before (including deletions): scores $F1 = 0.99$, confirming it understands the task and symbol set. On all 31 real scans, evaluated against the human-verified net-level ground truth, it

After: (none)

Change 336: delete | old pp. [8] | new pp. []

Before (including deletions): =

After: (none)

Change 337: replace | old pp. [8] | new pp. [9]

Before (including deletions): ($P = 0.97$, $R = 0.88$;

After: (precision 0.970, recall 0.880;

Change 338: replace | old pp. [8] | new pp. [9]

Before (including deletions): = 0.949). The VLM thus edges out deterministic join on aggregate (0.923 vs

After: 0.949), versus micro-F1

Change 339: replace | old pp. [8] | new pp. [9]

Before (including deletions): micro-F1), but

After: for

Change 340: replace | old pp. [8] | new pp. [9]

Before (including deletions): gap is not statistically significant: a paired bootstrap over the 31 images puts the difference at +0.033 with a 95% CI of $[-0.009, +0.078]$, which includes zero. Two further findings qualify the comparison. First, the VLM's accuracy degrades with circuit complexity: it

After: geometric join, and

Change 341: replace | old pp. [8] | new pp. [9]

Before (including deletions): ($F1 = 1.00$)

After: under the component-pair metric

Change 342: replace | old pp. [8] | new pp. [9]

Before (including deletions): 21

After: 21/31 images. The paired VLM-minus-join difference is +0.033, with 95% bootstrap CI $[-0.009, +0.078]$. Inclusion

Change 343: insert | old pp. [] | new pp. [9]

Before (including deletions): (none)

After: zero does not establish equivalence. These results do not establish comparative cost superiority or a global guarantee against incorrect merges for either method. Autonomous performance requires separately auditable evaluation;

Change 344: replace | old pp. [8] | new pp. [9]

Before (including deletions): 31 circuits, including all the small 3–5-component cases, but on the largest boards it systematically omits connections, with recall falling to 0.63– 0.68 on the three densest boards, exactly where a netlist error is most costly; its precision otherwise stays high (0.97 micro), so its failure mode is dropped nets rather than shorts. The macro-F1 (0.949) is correspondingly inflated by the many easy small circuits, which is why we report the pair-weighted micro-F1 as primary. Second, the cost and form of the output differ from the geometric pipeline: each image consumes on the order of 105 tokens, the result is free-form text rather than a structured netlist, and nothing in the model guarantees electrical validity (e.g. that a two-terminal component is

After: conditional scores do

Change 345: replace | old pp. [8] | new pp. [9]

Before (including deletions): shorted). Our geometric join is statistically indistinguishable from the VLM on these circuits while being deterministic, two to three orders of magnitude cheaper, structurally guaranteed against shorts by the self-loop guard, and simulatable by construction. We therefore treat the VLM as an upper reference for what a frontier model can do here, not as a connectivity module to ship: an explicit model reaches the same accuracy and also gives the determinism, low cost, and structural validity a deployed digitizer needs.

After: certify pin- level or simulation correctness.

Change 346: replace | old pp. [8] | new pp. [9]

Before (including deletions): endpoint-graph model resolves ambiguity that radius- based methods cannot: by representing both

After: endpoint graph represents

Change 347: replace | old pp. [8] | new pp. [9]

Before (including deletions): end- points

After: endpoints

Change 348: delete | old pp. [8] | new pp. []

Before (including deletions): as nodes

After: (none)

Change 349: replace | old pp. [8] | new pp. [9]

Before (including deletions): typed edges, it distinguishes between

After: separate rules for endpoint proximity, component assignment and mid-span contacts. Component-first assignment selects

Change 350: insert | old pp. [] | new pp. [9]

Before (including deletions): (none)

After: nearby component by OBB distance and then a pin by Euclidean distance. These geometric choices provide inspectable connectivity candidates, but do not establish that a nearby

Change 351: insert | old pp. [] | new pp. [9]

Before (including deletions): (none)

After: is electrically connected or

Change 352: delete | old pp. [8] | new pp. []

Before (including deletions): passes near

After: (none)

Change 353: replace | old pp. [8] | new pp. [9]

Before (including deletions): component (endpoint–wire–body edge) and one that connects to it (endpoint–pin edge with directional preference). The component-first assignment further disambiguates by using

After: selected pin has

Change 354: replace | old pp. [8] | new pp. [9]

Before (including deletions): geometric constraint that wires must reach a component's pin, not just its bounding box.

After: correct device-specific meaning.

Change 355: replace | old pp. [8] | new pp. [9]

Before (including deletions): addresses the fundamental limitation of detector noise: even a 97.6% accurate wire detector drops ~2% of wire endpoints, and those dropped endpoints are precisely the

After: proposes

Change 356: delete | old pp. [8] | new pp. []

Before (including deletions): that matter most

After: (none)

Change 357: replace | old pp. [8] | new pp. [9]

Before (including deletions): simulation accuracy. The b-matching formulation bounds the damage from mis-detection (each floating pin gets

After: pins left in single-component nets. Each floating-pin assignment selects

Change 358: replace | old pp. [8] | new pp. [9]

Before (including deletions): recovery edge), while

After: recovery target, and

Change 359: replace | old pp. [8] | new pp. [9]

Before (including deletions): self-loop

After: shared-component

Change 360: replace | old pp. [8] | new pp. [9]

Before (including deletions): prevents

After: can reject additional merges. Neither mechanism repairs shorts already present in

Change 361: replace | old pp. [8] | new pp. [9]

Before (including deletions): catastrophic short-circuit failures that plague unconstrained approaches.

After: base graph. Wire-detection F1 does not measure an endpoint-drop rate or establish the causal contribution of missing endpoints to simulation failures.

Change 362: insert | old pp. [] | new pp. [9]

Before (including deletions): (none)

After: Annotation was performed by a single human annotator using a purpose-built verification tool, and no per-image timing log was retained, so a per-image annotation-cost figure cannot be reported. A proposed expansion protocol would stratify sampling by component count, drafter, crossing/bus structure and capture conditions, use independent annotation followed by adjudication, and record per-image effort and provenance logs. This is a proposed expansion protocol, not completed annotation.

Change 363: replace | old pp. [8] | new pp. [9]

Before (including deletions): hand-drawn

After: hand-drawn

Change 364: replace | old pp. [8] | new pp. [9]

Before (including deletions): validated, as

After: validated. Table 7 distinguishes the evaluated connectivity scope from generic pin guesses and illustrative export. Listing a device group

Change 365: replace | old pp. [8] | new pp. [9]

Before (including deletions): extension

After: not establish separately detected model classes: the detector vocabulary merges multiple original classes. Pin construction depends on the component type supplied

Change 366: delete | old pp. [8] | new pp. []

Before (including deletions): circuits contain- ing devices outside

After: (none)

Change 367: replace | old pp. [8] | new pp. [9]

Before (including deletions): simulatable set (switches, transform- ers, etc.), which we exclude

After: netlist stage; a template for a named complex device does not recover that identity

Change 368: insert | old pp. [] | new pp. [9]

Before (including deletions): (none)

After: a merged detector label. Generic pin geometry is not evidence of validated device-specific pin relationships. Component detection has reported mAP@0.5 of 88.5%, with crossover recall of 70.7%. Geometric edge types 2, 4 and 5 do not consult crossover class labels when joining nearby wire fragments or mid-span contacts, creating a possible false-merge pathway. An audit using annotated component boxes found 13 crossover instances across eight benchmark images. In C242_D1_P1, suppressing two crossing-adjacent endpoint–endpoint edges changed component-pair TP/FP/FN from 27/4/0 to 19/0/8. Removing these edges also removes true connections; this is not a demonstrated mitigation gain. Detector-miss effects were not measured. This conditional joining intervention does not attribute

Change 369: replace | old pp. [8] | new pp. [9]

Before (including deletions): net-level benchmark.

After: four false pairs to missed or misclassified detector outputs and is not a prevalence estimate. Crossover-aware suppression remains an unvalidated intervention, and the impact of crossover recog- nition errors on autonomous netlists remains unresolved.

Change 370: delete | old pp. [8] | new pp. []

Before (including deletions): connectivity

After: (none)

Change 371: delete | old pp. [8] | new pp. []

Before (including deletions): frontier

After: (none)

Change 372: delete | old pp. [8] | new pp. []

Before (including deletions): the same

After: (none)

Change 373: replace | old pp. [8] | new pp. [9]

Before (including deletions): eval- uating

After: evaluating

Change 374: replace | old pp. [8] | new pp. [9]

Before (including deletions): fu- ture

After: future

Change 375: delete | old pp. [8] | new pp. []

Before (including deletions): Component detection at 88.5% mean average precision (mAP@0.5) also leaves room for improvement, particularly on crossovers (70.7% recall).

After: (none)

Change 376: replace | old pp. [8] | new pp. [9]

Before (including deletions): component- detection model

After: component detector

Change 377: replace | old pp. [8] | new pp. [9]

Before (including deletions): diverged markedly

After: differed

Change 378: replace | old pp. [8] | new pp. [9]

Before (including deletions): other

After: retained

Change 379: replace | old pp. [8] | new pp. [9]

Before (including deletions): this exclusion reduces the training set but avoids domain-shift artifacts from an outlier style. Finally, the

After: performance on that excluded style is not established here. The

Change 380: replace | old pp. [8] | new pp. [9]

Before (including deletions): a first-pass

After: an authored

Change 381: delete | old pp. [8] | new pp. []

Before (including deletions): of detector noise

After: (none)

Change 382: replace | old pp. [8] | new pp. [9]

Before (including deletions): calibrated fit; with the new real-image net- level metric available, calibrating it

After: cali- brated fit

Change 383: replace | old pp. [8] | new pp. [9]

Before (including deletions): the

After: detector errors. Calibration to

Change 384: replace | old pp. [8] | new pp. [9]

Before (including deletions): detector failure distribution

After: errors

Change 385: replace | old pp. [8] | new pp. [9]

Before (including deletions): a natural next step.

After: future work.

Change 386: replace | old pp. [8] | new pp. [9]

Before (including deletions): a complete deterministic pipeline for hand-drawn circuit digitization, built on

After: structural circuit-netlist recovery using

Change 387: replace | old pp. [8] | new pp. [9]

Before (including deletions): extractor that yields an endpoint representation

After: extractor

Change 388: delete | old pp. [8] | new pp. []

Before (including deletions): a two-stage

After: (none)

Change 389: replace | old pp. [8] | new pp. [9]

Before (including deletions): join, together with, to our knowledge, the first human-verified net-level connectivity benchmark for hand-drawn circuits. The endpoint-graph join

After: joining

Change 390: replace | old pp. [8] | new pp. [9]

Before (including deletions): completion

After: completion. On the 31-image within-corpus benchmark with annotated component boxes, component-pair micro-F1

Change 391: replace | old pp. [8] | new pp. [9]

Before (including deletions): most robust under synthetic error injection (F1 =

After: 0.8903; perfect-wire input gives 0.8898. These scores round alike but do not establish an autonomous detection bottleneck. The component-pair metric does not certify pin identities, exact net partitions, absence of shorts, or simulation equivalence. Synthetic L4 connectivity F1 is

Change 392: replace | old pp. [8] | new pp. [9]

Before (including deletions): at the highest severity vs.

After: versus

Change 393: replace | old pp. [8] | new pp. [9]

Before (including deletions): baseline), and, paired with a high-precision scale-relative base, attains connectivity micro-F1 = 0.890

After: baseline. The VLM comparison is conditional

Change 394: replace | old pp. [8] | new pp. [9]

Before (including deletions): real images against human-verified ground truth, beating the prior completion default (0.83)

After: annotated component boxes

Change 395: delete | old pp. [8] | new pp. []

Before (including deletions): every deterministic baseline, including the connected-component net-tracing recipe used by recent systems (0.62) and a Hough-plus-proximity line tracer (0.81), all on identical detected wires. Running the join on perfect wires

After: (none)

Change 396: replace | old pp. [8] | new pp. [9]

Before (including deletions): raise micro-F1 at all, showing the join, not wire detection,

After: establish equivalence or cost superiority. Export remains illustrative: values and device models require external specification, and real-scan simulation equivalence

Change 397: replace | old pp. [8, 9] | new pp. [9]

Before (including deletions): the bottleneck. A first-party experiment with a frontier VLM reaches 0.923 on the same images, a difference from our join that is not statistically significant (paired 95% CI [-0.009, +0.078]), but at two to three orders of magnitude 8 VOLUME 11, 2023 higher cost

After: unvalidated. Broader styles, dense buses,

Change 398: replace | old pp. [9] | new pp. [9]

Before (including deletions): without simulatable, structurally valid output, supporting the explicit geometric design. Future work includes scaling the human-verified benchmark, calibrating the synthetic error model to the measured detector distribution, and extending coverage to devices outside the simulatable set.

After: mixed-scale schematics require future evaluation.

Change 399: insert | old pp. [] | new pp. [9, 10]

Before (including deletions): (none)

After: VOLUME 11, 2023 9 TABLE 7. Connectivity evaluation, generic pin construction, and illustrative export are distinct capabilities. Device group Evaluated scope Pin construction Export limitation R/C/L/D/Q, voltage sources and ICs Component-pair connectivity on the benchmark; no universal validation of pin templates Geometric templates; endpoint clustering for eligible discrete/source types, generic layouts for ICs Illustrative defaults/substitutions; values and device models require external specification. ICs are skipped because no export model is implemented. Switches Absent from the evaluated electrical subset; switch-specific connectivity unvalidated Two pins from the shorter OBB-edge midpoints; long-axis AABB fallback Fixed 0.001 Ω resistor if pins 0 and 1 map to distinct nodes; otherwise no switch line. No switch state or control model is recovered. Transformers, thyristors, optocouplers and other complex devices Device-specific connectivity unvalidated Named transformer type receives a two-pin geometric guess; optocoupler template and unknown-type fallback also exist These complex types have no implemented export model and are skipped; geometric guesses do not establish winding, control or isolation semantics. Metric boundary (all evaluated groups) Component pairs discard terminal identity F1 does not certify pin assignment or the exact net partition F1 does not certify absence of shorts or simulation equivalence.

Change 400: replace | old pp. [9] | new pp. [10]

Before (including deletions): IV-F,

After: IV-G,

Change 401: replace | old pp. [9] | new pp. [10]

Before (including deletions): available

After: available

Change 402: replace | old pp. [9] | new pp. [10]

Before (including deletions): <https://github.com/boscochanam/circuit-digitization>.

After: <https://github.com/boscochanam/circuit-digitization>, including the per-image to CGHD drafter mapping used for the per-drafter analysis (Table 5).

Change 403: replace | old pp. [9] | new pp. [10]

Before (including deletions): <https://huggingface.co/boscochanam/circuit-component-detector>,

After: <https://huggingface.co/boscochanam/circuit-component-detector>,

Change 404: insert | old pp. [] | new pp. [10]

Before (including deletions): (none)

After: S. Gao et al., “A memristor-based neural network circuit with classical conditioning and fear generalization,” IEEE Trans. Consumer Electron., vol. 72, no. 2, pp. 3213–3224, May 2026, doi: 10.1109/TCE.2026.3663359. [7] S. Gao et al., “Biologically plausible memristive decision-making circuit for adaptive control in industrial autonomous navigation,” IEEE Trans. Ind. Informat., vol. 22, no. 6, pp. 4801–4812, Jun. 2026, doi: 10.1109/TII.2026.3668794. [8]

Change 405: replace | old pp. [9] | new pp. [10]

Before (including deletions): [7]

After: [9]

Change 406: replace | old pp. [9] | new pp. [10]

Before (including deletions): [8]

After: [10]

Change 407: replace | old pp. [9] | new pp. [10]

Before (including deletions): [9]

After: [11]

Change 408: replace | old pp. [9] | new pp. [10]

Before (including deletions): [10]

After: [12]

Change 409: replace | old pp. [9] | new pp. [10]

Before (including deletions): [11]

After: [13]

Change 410: replace | old pp. [9] | new pp. [10]

Before (including deletions): [12]

After: [14]

Change 411: replace | old pp. [9] | new pp. [10]

Before (including deletions): [13]

After: [15]

Change 412: replace | old pp. [9] | new pp. [10]

Before (including deletions): [14]

After: [16]

Change 413: replace | old pp. [9] | new pp. [10]

Before (including deletions): [15]

After: [17]

Change 414: replace | old pp. [9] | new pp. [10]

Before (including deletions): [16]

After: [18]

Change 415: replace | old pp. [9] | new pp. [10]

Before (including deletions): [17]

After: [19]

Change 416: replace | old pp. [9] | new pp. [10]

Before (including deletions): [18]

After: [20]

Change 417: replace | old pp. [9] | new pp. [10]

Before (including deletions): [19]

After: [21]

Change 418: replace | old pp. [9] | new pp. [10]

Before (including deletions): [20]

After: [22]

Change 419: replace | old pp. [9] | new pp. [10]

Before (including deletions): [21]

After: [23]

Change 420: replace | old pp. [9] | new pp. [10]

Before (including deletions): [22]

After: [24]

Change 421: replace | old pp. [9] | new pp. [10]

Before (including deletions): [23]

After: [25]

Change 422: replace | old pp. [9] | new pp. [10]

Before (including deletions): [24]

After: [26]

Change 423: insert | old pp. [] | new pp. [10]

Before (including deletions): (none)

After: 10 VOLUME 11, 2023

Change 424: replace | old pp. [9] | new pp. [11]

Before (including deletions): [25]

After: [27]

Change 425: replace | old pp. [9] | new pp. [11]

Before (including deletions): [26]

After: [28]

Change 426: delete | old pp. [9] | new pp. []

Before (including deletions): VOLUME 11, 2023 9

After: (none)

Change 427: delete | old pp. [10] | new pp. []

Before (including deletions): 10

After: (none)

Change 428: insert | old pp. [] | new pp. [11]

Before (including deletions): (none)

After: 11